

Worksheet 1

Deconstructing Plots

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1 Some example datasets

In this workshop, we'll be using two common example datasets to illustrate data visualizations: the `mtcars` and `diamonds` datasets. Both are supplied by any standard installation of the R language.

The `mtcars` dataset is a common small example dataset used for illustrating different plots. It contains the technical specifications of different car models by various manufacturers. Some lines of the dataset are given below, as it may be helpful to understanding some of the plots below.

```
mtcars[1:10,]
```

##		mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb	brand
##	Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4	Mazda
##	Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4	Mazda
##	Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1	Datsun
##	Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1	Hornet
##	Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2	Hornet
##	Valiant	18.1	6	225.0	105	2.76	3.460	20.22	1	0	3	1	Valiant
##	Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0	0	3	4	Duster
##	Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	0	4	2	Merc
##	Merc 230	22.8	4	140.8	95	3.92	3.150	22.90	1	0	4	2	Merc
##	Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	0	4	4	Merc

The `diamonds` dataset is another common example dataset. It is much bigger than the `mtcars` dataset. It contains data on the properties and auction price of many diamonds. Some rows are again

```
diamonds[1:10,]
```

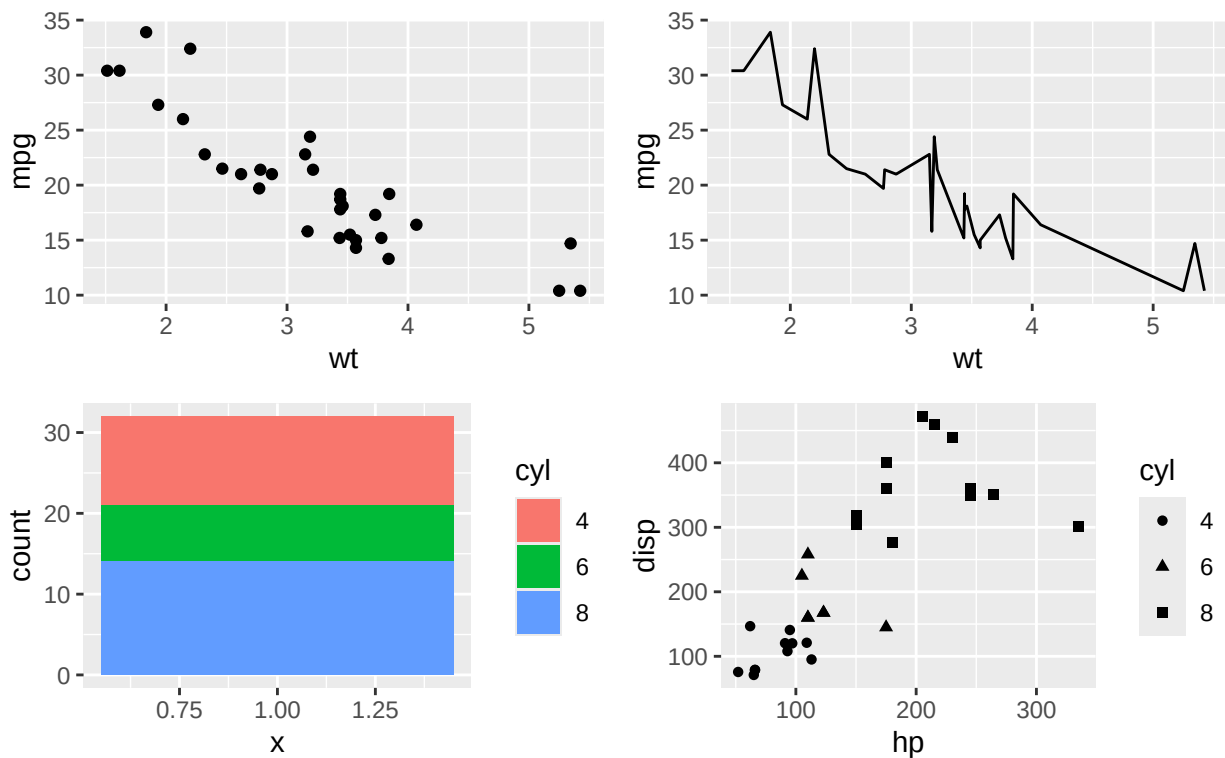
```
## # A tibble: 10 x 10
```

##	carat	cut	color	clarity	depth	table	price	x	y	z	
##	<dbl>	<ord>	<ord>	<ord>	<dbl>	<dbl>	<int>	<dbl>	<dbl>	<dbl>	
##	1	0.23	Ideal	E	SI2	61.5	55	326	3.95	3.98	2.43
##	2	0.21	Premium	E	SI1	59.8	61	326	3.89	3.84	2.31
##	3	0.23	Good	E	VS1	56.9	65	327	4.05	4.07	2.31
##	4	0.29	Premium	I	VS2	62.4	58	334	4.2	4.23	2.63
##	5	0.31	Good	J	SI2	63.3	58	335	4.34	4.35	2.75
##	6	0.24	Very Good	J	VVS2	62.8	57	336	3.94	3.96	2.48
##	7	0.24	Very Good	I	VVS1	62.3	57	336	3.95	3.98	2.47
##	8	0.26	Very Good	H	SI1	61.9	55	337	4.07	4.11	2.53
##	9	0.22	Fair	E	VS2	65.1	61	337	3.87	3.78	2.49
##	10	0.23	Very Good	H	VS1	59.4	61	338	4	4.05	2.39

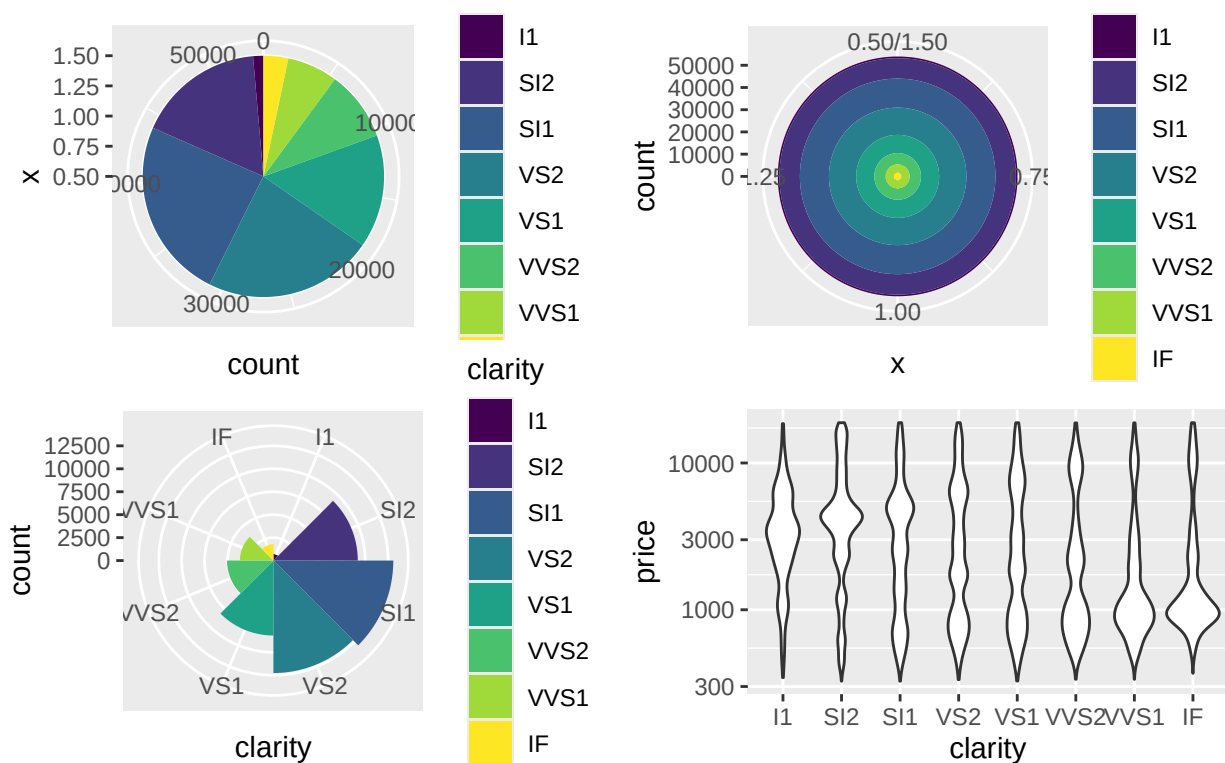
2 Finding the components of a plot

Grab a partner, and take a look at the following plots, produced from a common example dataset containing the technical specifications of different car models. List the visual primitives each plot is composed of, and which variables of the dataset are displayed in the plot. Investigate which messages they communicate to a reader.

2.1 mtcars plots



2.2 diamonds plots



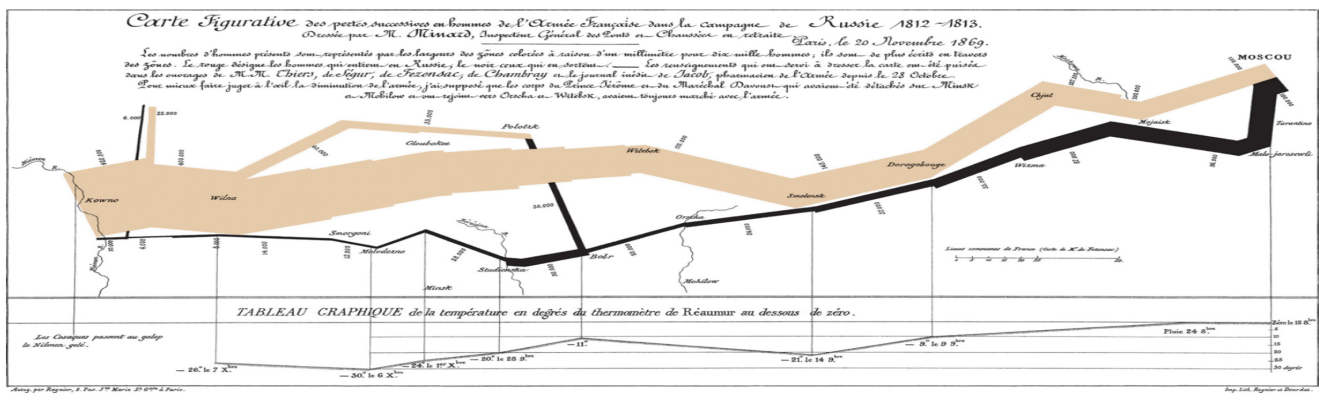
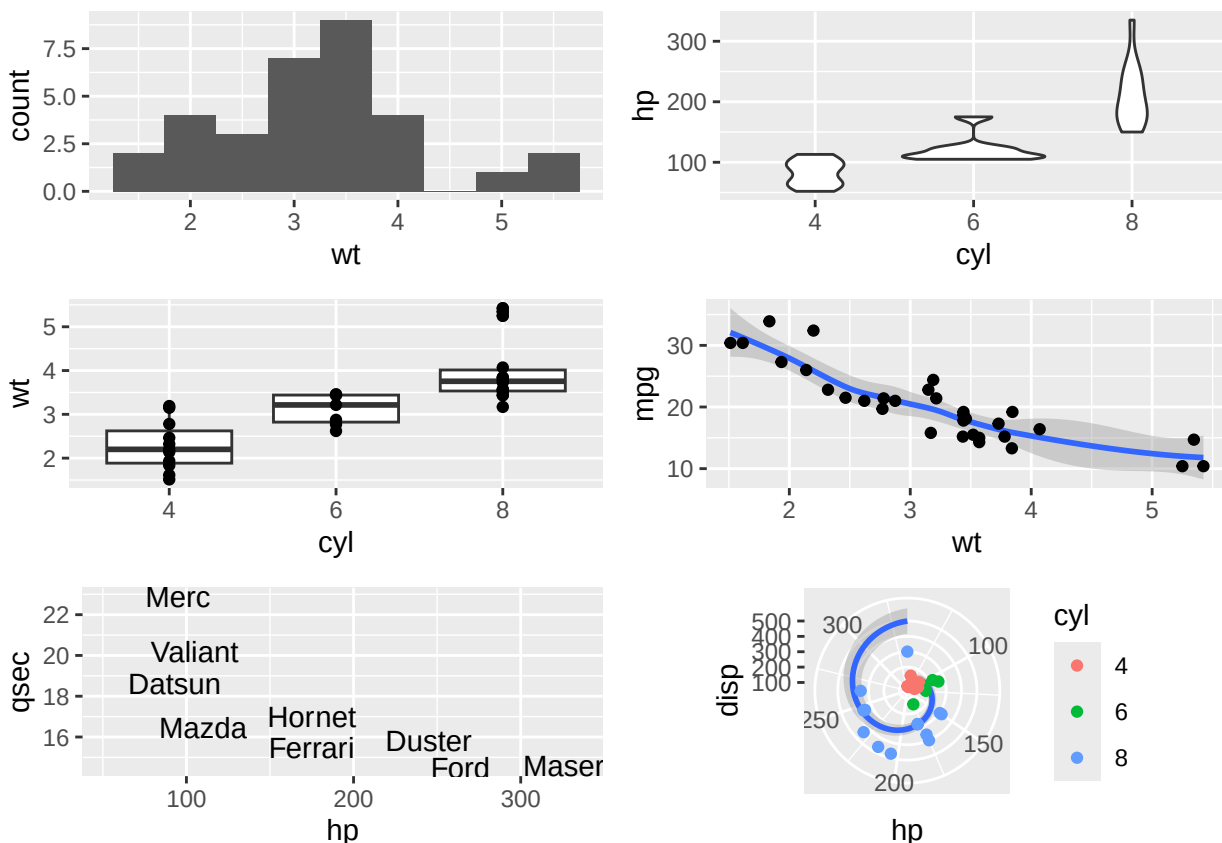


Figure 1: Plot of napoleons troop movement in the winter of 1812-13.

3 Statistics in plotting

Have a look at the plots below with your partner. Which statistics are computed from the dataset to make the plot, and which graphical components do they correspond to? Does this assignment make sense?

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```



4 Napoleons plot in the grammar of graphics

Fig. 1 is a reproduction of a famous 19th century plot of Napoleons 1812-13 winter campaign by Charles Minard. This plot was originally drawn by hand and distributed by lithography; nonetheless, it can be described using the grammar of graphics. Find the components of this graph, the statistics computed on the raw data (number of troops and location in Napoleons army at different times, weather data) and how they relate to each other. Which other features are present in the graph, and how does it deviate from the grammar of graphics?